Project Objective:

The goal behind this project was to design two PID controllers that could control the X and Y axes of a plate hooked up to the PLC, and control the plate into balancing the ball, making it drift to several set quadrants on the plate before bringing it back to center using the sequential block logic to give it continuous directions.

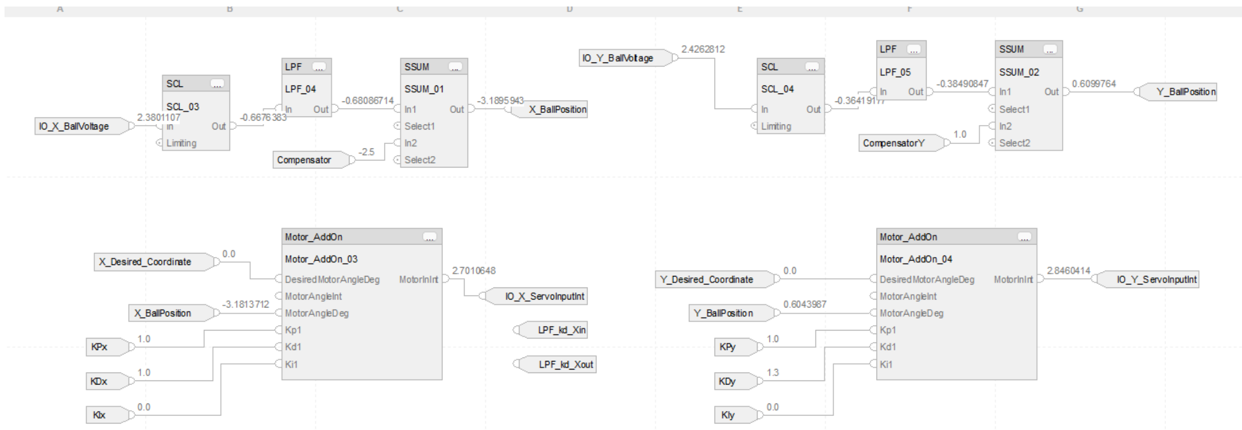

Figure 1. PID function block logic for controlling the ball coordinates

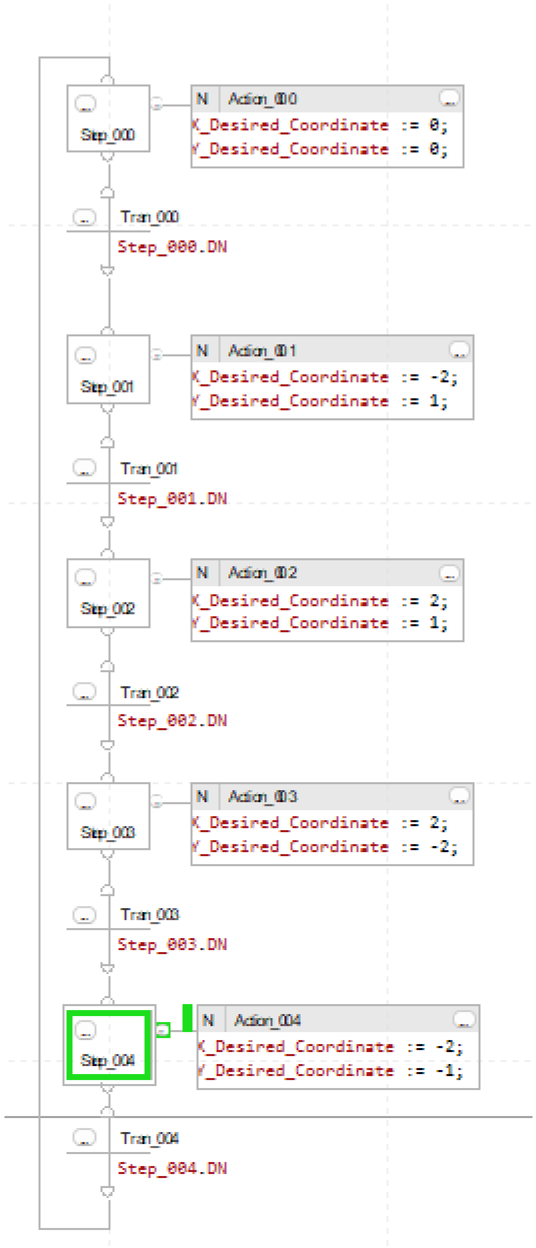

Figure 2. Sequential Logic Chart for balancing ball in different sectors in sequence

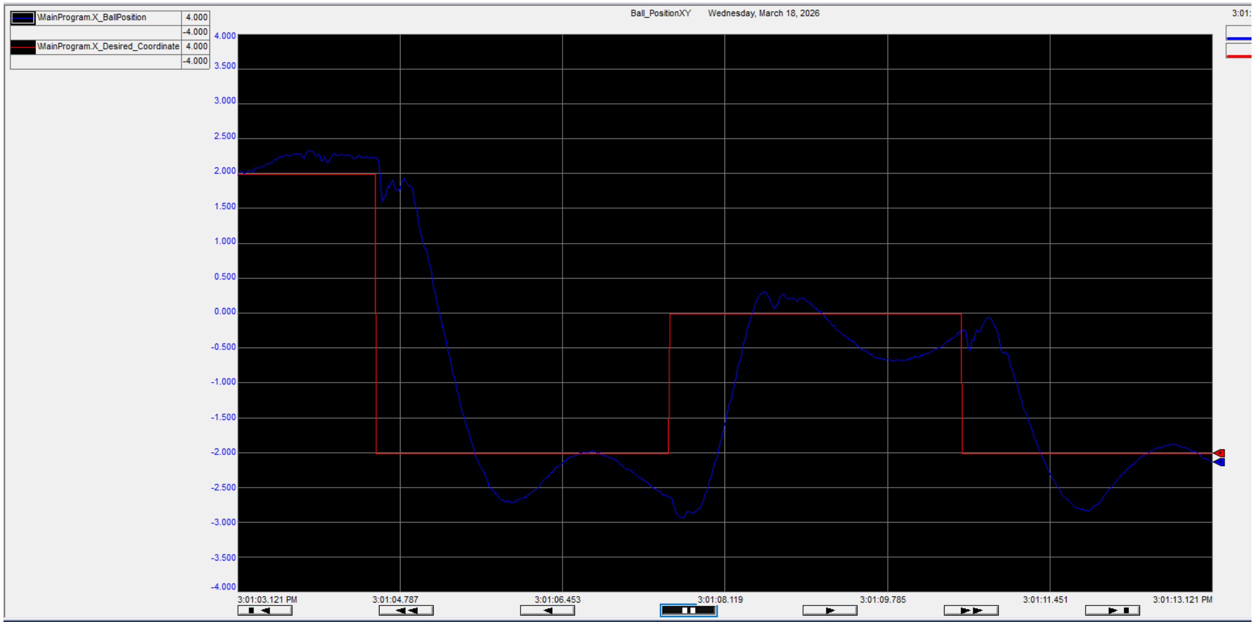

Figure 3. Trend of low-pass filter smoothing effect on controller output

Table 1: Time-domain parameters of PID controller

Measurement	Axis	Value
Percent Overshoot (Mp)	X Axis	7 %
	Y Axis	11 %
10–90% Rise Time (Tr)	X Axis	0.55 s
	Y Axis	0.62 s
Settling Time (Ts)	X Axis	1.8 s
	Y Axis	2.1 s
Steady-State Error (Ess)	X Axis	0.005 m (≈ 0.5cm)
	Y Axis	0.007 m (≈ 0.7 cm)

$Tsequence$ = 19.8 s

Technical Summary

The ball and plate lab was one of my more difficult projects, but I learned many practical skills when it comes to designing PIDs and measuring their effects through different filters and the PLC machine itself. My biggest roadblock was finding the right amount of voltage to properly allow the plate to follow my constraints, but after I did it functioned sufficiently and inside all the restraints. I had first set up the function block with my past PID setups and loops, and then duplicated them and applied them to the separate axes. Then I set up a sequential logic block that would set up a sequence in which coordinates would be visited first by the rolling ball before returning back to the center.

Though it took awhile, it helped me notice granular details that are often missed in theory, like seeing the real effect of what adding the compensator did to the offset, as well as how to improve loop stability. I also learned how to optimize a low pass filter to cleanly send my inputs to the plate. Learning to balance a ball on a plate is one of the most important tests of understanding control theory, and will come in handy in future careers like learning aerospace stabilization and other motion controlled systems.